

# Locating Zeros and Turning-Point Constraints

## Answer Key

### Precalculus

#### Quick Answers

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|------------------------------------------------------|----------------------------------------------------|
| 1. 3, 4                                              | 7. (a) degree = 5, (b) turning-point bound = 4     |
| 2. 5                                                 | 8. 2, -2, 3, -3                                    |
| 3. (a) zeros = $\frac{1}{2}$ , -4, —                 | 9. (a) turning-point bound = 4, —                  |
| 4. 0, 4, -1                                          | 10. (a) factored form = $(x - 3)(2x - 1)(x + 2)$ , |
| 5. (a) turning-point bound = 3, (b) limit = $\infty$ | (b) zeros = 3, $\frac{1}{2}$ , -2                  |
| 6. (a) factored form = $x(x + 3)(x - 2)$ , (b)       | 11. (a) zeros = 0, 2, -2, —                        |
| zeros = 0, -3, 2                                     | 12. (a) leading coefficient = 2, —                 |
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#### What is verified

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16 of 20 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 3, 9, 11, 12. The worked solution states what a correct response must contain.

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#### Curriculum

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**Level:** Precalculus

HSA-APR.B.3 – problems 1, 3, 4, 6, 8, 10, 11, 12

HSF-IF.C – problems 2, 5, 7, 9, 12

*Difficulty 1–5 of 5 across 20 tagged checks.*

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1. Find all real zeros of  $p(x) = x^2 - 7x + 12$ .

**Solution:** Two numbers multiplying to 12 and adding to -7 are -3 and -4, so  $p(x) = (x - 3)(x - 4)$ .

$$\boxed{x = 3 \quad \text{and} \quad x = 4}$$

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2. A polynomial has degree 6. Greatest possible number of turning points?

**Solution:** A polynomial of degree  $n$  has at most  $n - 1$  turning points, because its derivative has degree  $n - 1$  and so at most  $n - 1$  roots. Here  $6 - 1 = 5$ .  $\boxed{5}$

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3.  $p(x) = (2x - 1)^2(x + 4)$ . (a) Zeros. (b) Multiplicity and behaviour.

**Solution (a):**  $2x - 1 = 0$  gives  $x = \frac{1}{2}$ ;  $x + 4 = 0$  gives  $x = -4$ .

$$x = \frac{1}{2} \quad \text{and} \quad x = -4$$

**Solution (b):** *Instructor-judged.* The zero at  $\frac{1}{2}$  comes from a squared factor, so it has multiplicity 2 and the curve touches the axis and turns back there. The zero at  $-4$  has multiplicity 1, so the curve crosses. A full answer gives both multiplicities and the matching behaviour; a strong one notes the multiplicities sum to 3, which is the degree.

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**4.** Find all real zeros of  $p(x) = x^3 - 3x^2 - 4x$ .

**Solution:** Common factor first, then factor the quadratic:

$$x^3 - 3x^2 - 4x = x(x^2 - 3x - 4) = x(x - 4)(x + 1)$$

Pulling out  $x$  is what keeps the zero at 0 visible.

$$x = -1, \quad x = 0, \quad x = 4$$

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**5.**  $p(x) = x^4 - 3x^2 + 1$ . **(a)** Turning-point bound. **(b)** Limit at  $-\infty$ .

**Solution (a):** Degree 4, so at most  $4 - 1 = 3$  turning points.

3

**Solution (b):** The degree is even and the leading coefficient positive, so both ends rise.

$\lim_{x \rightarrow -\infty} p(x) = \infty$

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**6.**  $p(x) = x^3 + x^2 - 6x$ . **(a)** Factor. **(b)** Zeros.

**Solution (a):**  $x^3 + x^2 - 6x = x(x^2 + x - 6)$ , and  $x^2 + x - 6 = (x + 3)(x - 2)$ .

$p(x) = x(x + 3)(x - 2)$

**Solution (b):** Each factor is linear, so all three zeros are simple.

$$x = -3, \quad x = 0, \quad x = 2$$

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**7.**  $p(x) = (x + 1)^2(x - 3)^3$ . **(a)** Degree. **(b)** Turning-point bound.

**Solution (a):** The degree is the sum of the multiplicities:  $2 + 3 = 5$ . Expanding is unnecessary.

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**Solution (b):** At most  $5 - 1 = 4$  turning points.

4

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**8.** Find all real zeros of  $p(x) = x^4 - 13x^2 + 36$ .

**Solution:** It is a quadratic in  $x^2$ :  $(x^2 - 4)(x^2 - 9)$ , and each bracket is a difference of squares, giving  $(x - 2)(x + 2)(x - 3)(x + 3)$ .

$x = -3, \quad x = -2, \quad x = 2, \quad x = 3$

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- 9.** Degree 5, positive leading coefficient, three simple real zeros. **(a)** Turning-point bound. **(b)** Why at least two.

**Solution (a):**  $5 - 1 = 4$ .

4

**Solution (b):** *Instructor-judged.* Between two consecutive simple zeros the curve leaves the axis on one side and has to return to it, so it must turn at least once in each of the two gaps between the three zeros — at least two turning points. A correct answer makes that gap argument (or the equivalent sign-change argument) rather than simply asserting the number. A student who adds that the true count must be 2 or 4, matching the parity of the bound, has gone beyond the question and should be credited.

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- 10.**  $p(x) = 2x^3 - 3x^2 - 11x + 6$  with  $p(3) = 0$ . **(a)** Factor. **(b)** Zeros.

**Solution (a):** Dividing by  $x - 3$  leaves  $2x^2 + 3x - 2$ , which factors as  $(2x - 1)(x + 2)$ .

$p(x) = (x - 3)(2x - 1)(x + 2)$

**Solution (b):** Setting each factor to zero:  $x = 3$ ,  $x = \frac{1}{2}$ ,  $x = -2$ . The fractional zero is the one a guess-the-integer search never finds — the factor  $2x - 1$  is where it comes from.

$x = -2, \quad x = \frac{1}{2}, \quad x = 3$

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- 11.**  $p(x) = x^5 - 4x^3$ . **(a)** Factor and list zeros. **(b)** Multiplicities.

**Solution (a):**  $x^5 - 4x^3 = x^3(x^2 - 4) = x^3(x - 2)(x + 2)$ .

$x = -2, \quad x = 0, \quad x = 2$

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**Solution (b):** *Instructor-judged.* The zero at 0 has multiplicity 3; the zeros at 2 and  $-2$  have multiplicity 1. Every multiplicity is odd, so the curve crosses at all three — a repeated factor does not automatically mean a touch, and that is the point of the item. Credit an answer that also notes  $3 + 1 + 1 = 5$ , the degree.

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- 12.**  $p(x) = a(x + 2)^2(x - 1)(x - 3)$  with  $p(0) = 24$ . **(a)** Find  $a$ . **(b)** Turning points.

**Solution (a):** Substitute  $x = 0$ :

$$\begin{aligned} a(2)^2(-1)(-3) &= 24 \\ 12a &= 24 \end{aligned}$$

So

$a = 2$

**Solution (b):** *Instructor-judged.* The degree is 4, so the bound is 3 turning points. All three actually occur: the curve touches and turns at  $-2$ , then must turn once between  $-2$  and 1 and once between 1 and 3 in order to cross at both. A correct answer states the bound and argues the actual count; giving only the bound answers half the question.